OMEX Experiment Report — vcornea_sls_sweep_20261021_115344

3 combinations | 5, 5, 5 replicate(s) each | SimTime 30 days | injury at day 3.0

Experiment Archive

This report describes the experiment-level archive **vcornea_sls_sweep_scan.omex** (234,239 bytes) — the SED-ML/COMBINE description of the whole scan. The per-combination reports describe each combination's per-replicate archives; this one sits above them and compares the combos.

Parameter scan	SLS_Concentration ∈ [1500.0, 2000.0, 2500.0]
Scan points	3
Replicates / point	5
Total simulations	15

Experiment OMEX Contents

File	Size (B)
combo_data/vcornea_sls_sweep_combo001_cell_count.csv	19,384
combo_data/vcornea_sls_sweep_combo001_surfactant.csv	170,128
combo_data/vcornea_sls_sweep_combo001_thickness.csv	18,096
combo_data/vcornea_sls_sweep_combo002_cell_count.csv	19,361
combo_data/vcornea_sls_sweep_combo002_surfactant.csv	170,033
combo_data/vcornea_sls_sweep_combo002_thickness.csv	18,035
combo_data/vcornea_sls_sweep_combo003_cell_count.csv	19,338
combo_data/vcornea_sls_sweep_combo003_surfactant.csv	169,886
combo_data/vcornea_sls_sweep_combo003_thickness.csv	18,067
manifest.xml	1,662
metadata.rdf	1,586
scan.sedml	29,595
scan_metadata.json	1,360

Reading the SED-ML (comparison)

scan.sedml encodes the scan as a repeatedTask AND compares the combinations: 9 dataDescriptions (one per combo × metric, bundled under combo_data/) feed 6 plot2D outputs — each overlays the combinations for one signal — and a 18-series report.

plot2D (signal across combinations)	curves (= combinations)
Superficial (cell_count) across combinations	3
Wing (cell_count) across combinations	3
Basal (cell_count) across combinations	3
Stem (cell_count) across combinations	3
Total Amount (surfactant) across combinations	3
Height (thickness) across combinations	3

Provenance (OMEX Metadata)

Field	Value
Creator (experimenter)	Joel Vanin
Created	2026-10-21T18:38:29
Archived	2026-10-21T18:38:29
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Vanin et al. (2025). V-Cornea: a computational model of the corneal epithelium.
Keywords	CompuCell3D, cornea, parameter scan, agent-based model, wound healing

From metadata.rdf in the scan archive: attribution / provenance, not authentication.

Validation of the Experiment Archive

Layer	Result
Built-in SED-ML structure	PASS
Built-in OMEX manifest	PASS
biosimulators-utils (COMBINE)	PASS (0 errors, 1 warning group)

Combinations

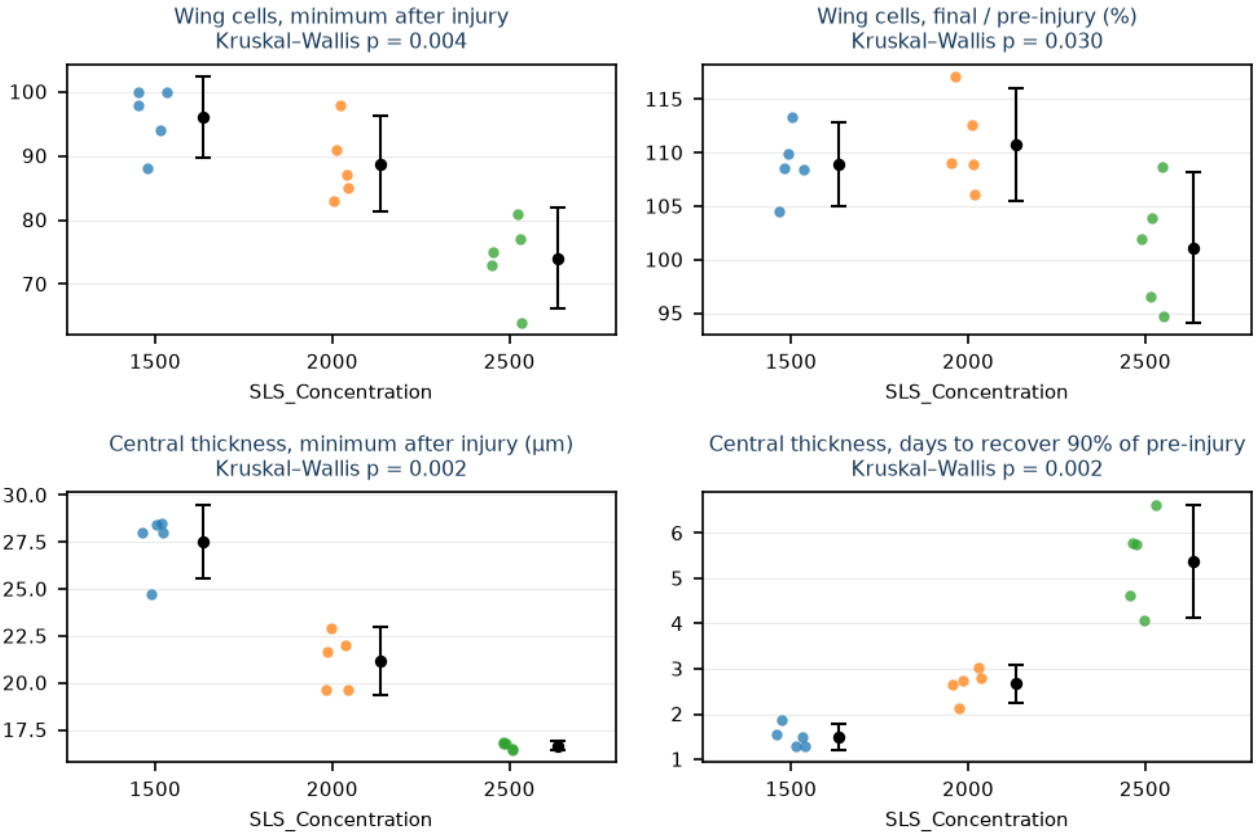
Combination	Label	Replicates	Unsuccessful
vcornea_sls_sweep_combo001	SLS_Concentration=1500.0	5	0
vcornea_sls_sweep_combo002	SLS_Concentration=2000.0	5	0
vcornea_sls_sweep_combo003	SLS_Concentration=2500.0	5	0

Summary Statistics (per-replicate endpoints)

Each endpoint is computed for every replicate separately, then summarised across the replicates of a combination as mean ± SD with the 95% confidence interval of the mean (t distribution) below it. The last column is a Kruskal–Wallis test of whether the endpoint differs across the combinations.

Endpoint / SLS_Concentration	1500 (n=5)	2000 (n=5)	2500 (n=5)	Kruskal–Wallis p
Wing cells, pre-injury mean	117.7 ± 2.2 [114.9, 120.4]	118.0 ± 3.2 [114.0, 121.9]	116.8 ± 2.7 [113.4, 120.1]	0.914
Wing cells, minimum after injury	96.0 ± 5.1 [89.7, 102.3]	88.8 ± 5.9 [81.4, 96.2]	74.0 ± 6.3 [66.1, 81.9]	0.004
Wing cells, final / pre-injury (%)	109.0 ± 3.1 [105.1, 112.9]	110.7 ± 4.2 [105.5, 116.0]	101.1 ± 5.6 [94.1, 108.1]	0.030
Central thickness, pre-injury mean (µm)	38.5 ± 0.7 [37.6, 39.5]	38.3 ± 0.6 [37.6, 39.1]	38.5 ± 0.9 [37.3, 39.6]	0.970
Central thickness, minimum after injury (µm)	27.5 ± 1.6 [25.6, 29.5]	21.2 ± 1.5 [19.3, 23.0]	16.7 ± 0.2 [16.4, 16.9]	0.002
Central thickness, final (µm)	43.0 ± 0.4 [42.5, 43.6]	42.4 ± 0.7 [41.5, 43.3]	35.1 ± 2.6 [31.8, 38.3]	0.006
Central thickness, days to recover 90% of pre-injury	1.5 ± 0.2 [1.2, 1.8]	2.7 ± 0.3 [2.3, 3.1]	5.4 ± 1.0 [4.1, 6.6]	0.002
Surfactant clearance half-life (h)	29.7 ± 0.8 [28.7, 30.6]	33.7 ± 0.4 [33.1, 34.2]	37.9 ± 0.4 [37.4, 38.4]	0.002

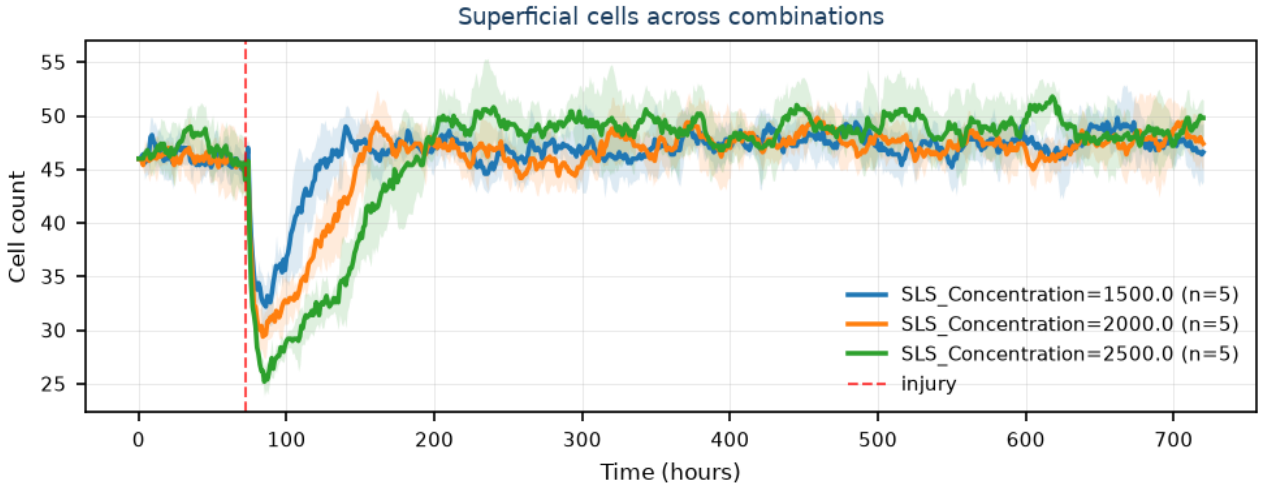
Pre-injury = before the injury time; minimum after injury = damage depth. Central thickness = mean over bins 4–9 (µm, the GUI's analysis; a bin with no epithelium left counts as 0). Surfactant half-life: hours from the applied peak until the total amount is down to half of it. Days to recover: from injury until central thickness, after its minimum, is back to 90% of its pre-injury mean (0 = never fell below 90%; replicates that did not recover within the run are left out and the remaining n is shown). With few replicates per combination the tests have little power and are not corrected for the number of endpoints; read them as descriptive.



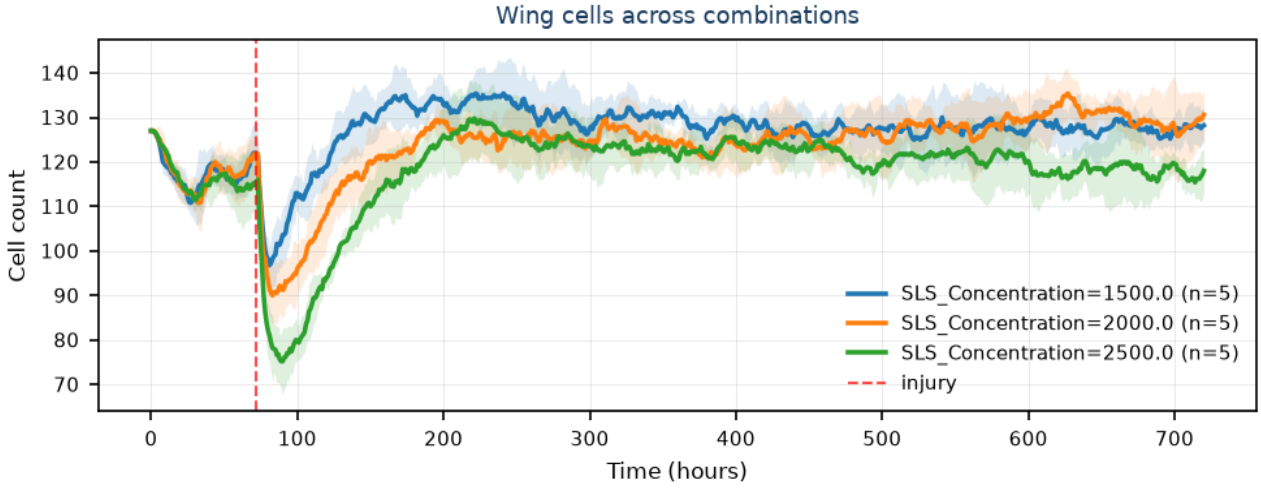
Key endpoints by combination: each dot is one replicate; the black point and bar are the mean and its 95% confidence interval.

Metric Comparisons (one line per combination)

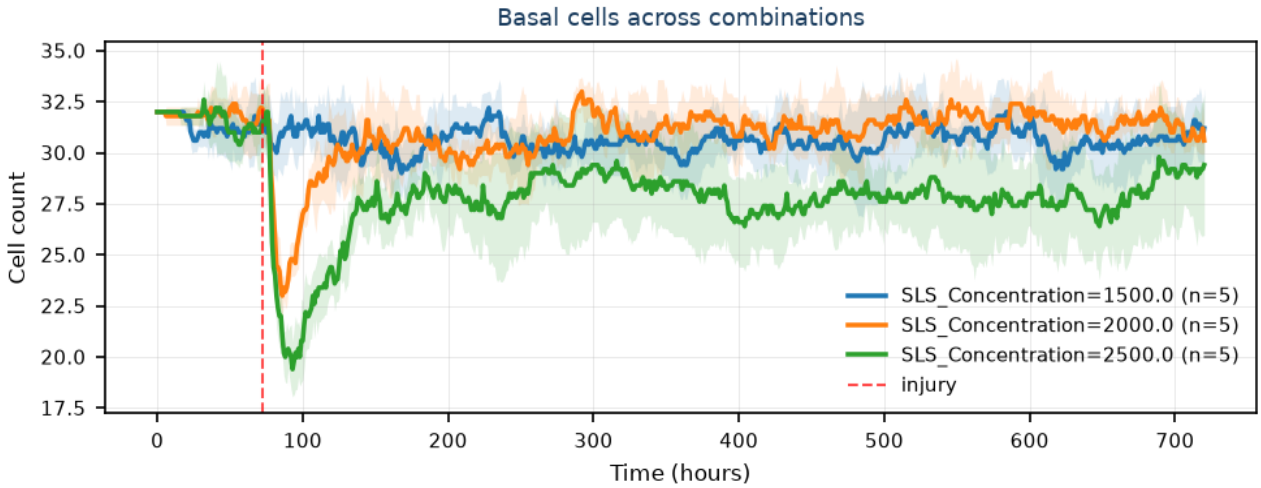
Each line is a combination's across-replicate mean and the shaded band ±1 SD across its replicates, read from the per-run OMEX archives. The legend gives the number of replicates n. Time in hours (thickness: days); the dashed red line marks injury.



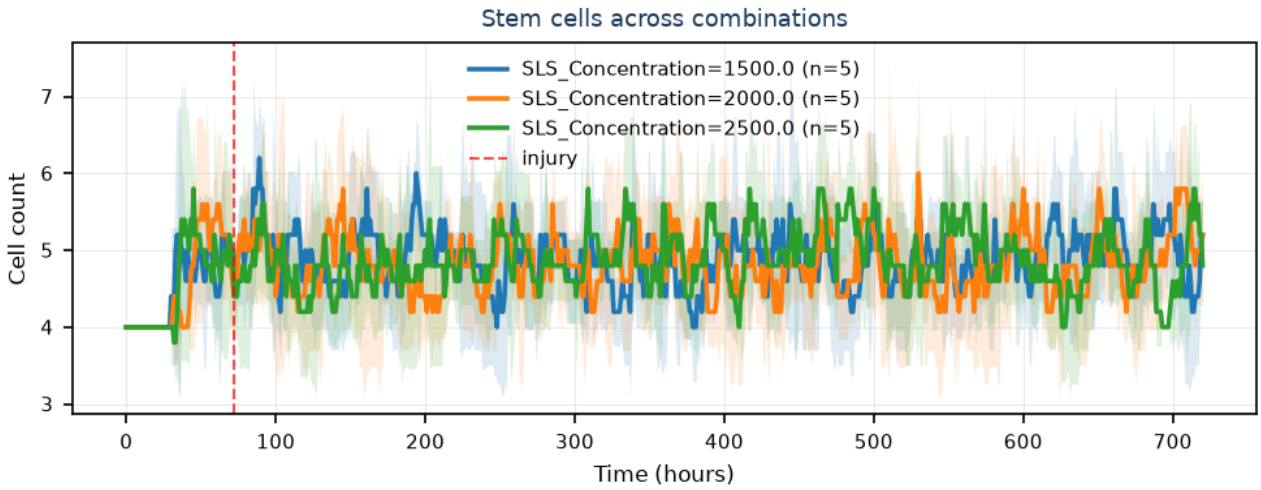
Superficial: replicate mean per combination vs Time, shaded ±1 SD across replicates



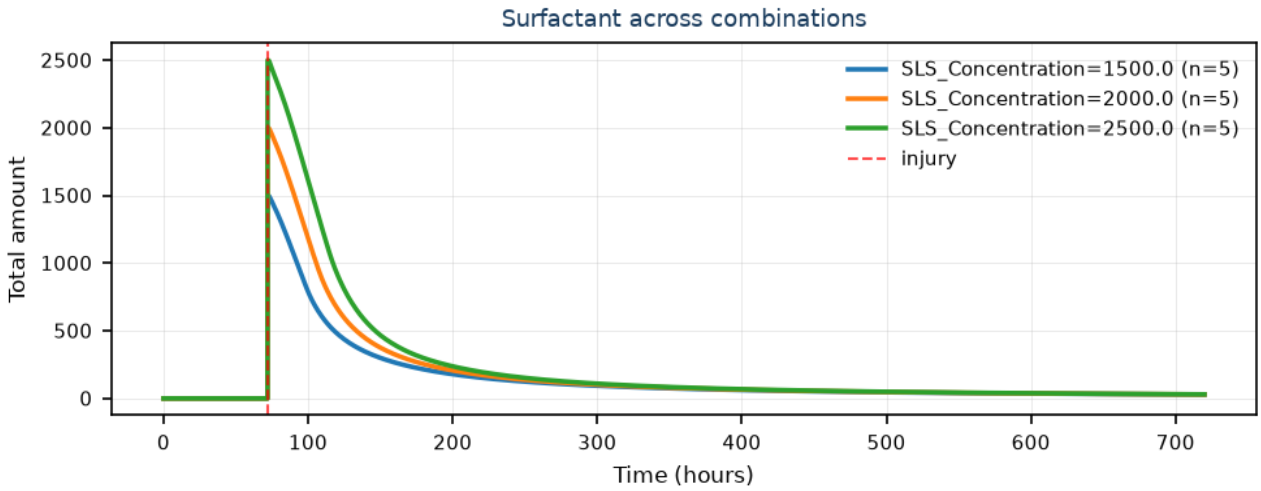
Wing: replicate mean per combination vs Time, shaded ±1 SD across replicates



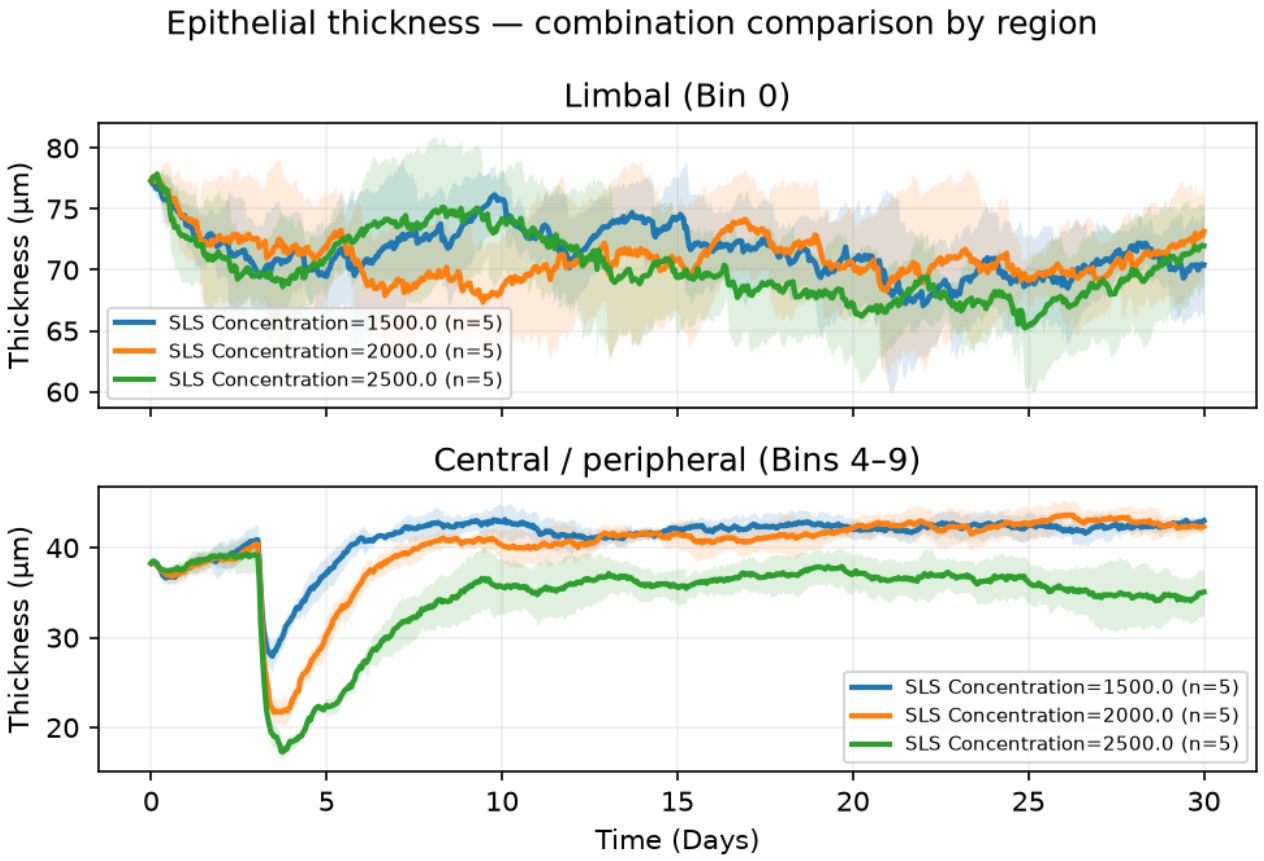
Basal: replicate mean per combination vs Time, shaded ±1 SD across replicates



Stem: replicate mean per combination vs Time, shaded ±1 SD across replicates



Surfactant Total Amount: replicate mean per combination vs Time, shaded ±1 SD across replicates

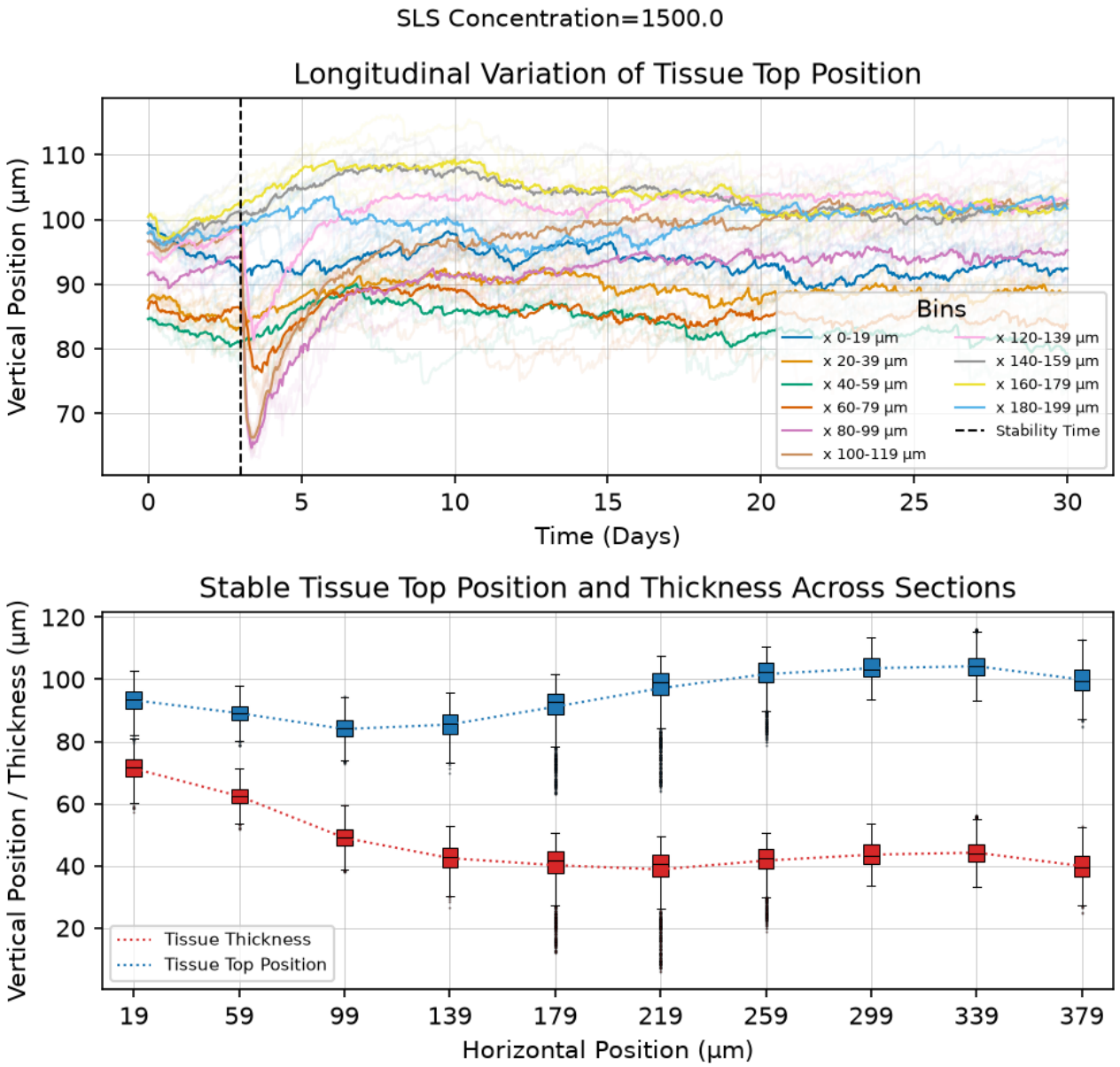


Epithelial thickness in µm over time in the limbus (bin 0) and the central/peripheral cornea (bins 4–9), one line per combination (replicate mean, shaded ±1 SD across replicates), drawn by the V-Cornea GUI's own analysis (Analyze Results → Thickness, comparison view). Thickness = (top position − membrane baseline) × 2 µm/voxel. Time in days.

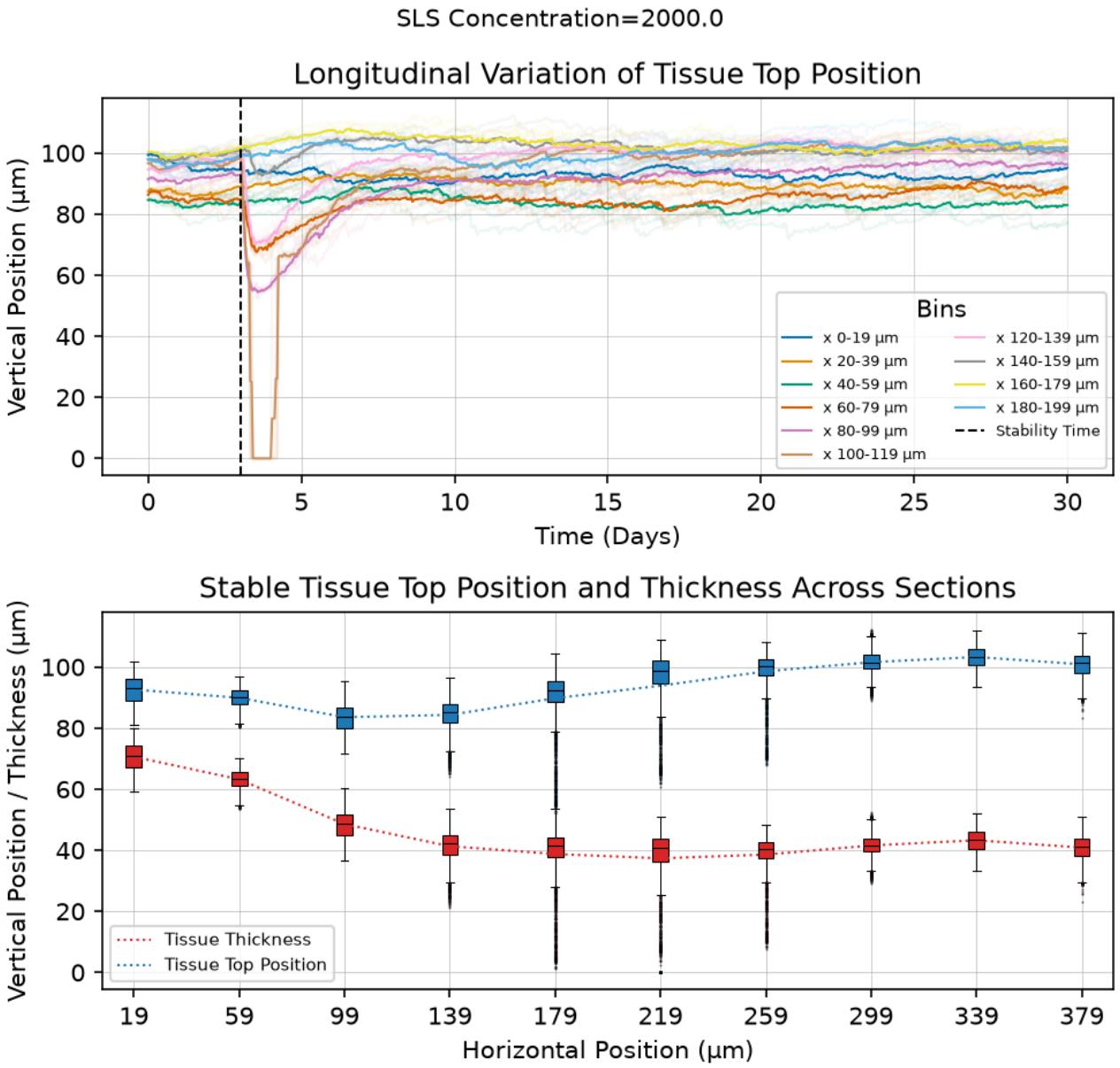
Tissue Thickness by Combination (paper view)

For each combination, the V-Cornea GUI's paper thickness figure (Analyze Results → Thickness, paper view; Vanin et al. 2025, Fig 8). Top: tissue top position in each of the 10 X-bins over time; faint lines are the replicates, solid lines their mean. Bottom: from the stability time (day 3) on, top position (blue) and thickness (red) across X, as box plots over replicates and time; thickness = (top position – membrane baseline) × 2 µm/voxel.

SLS_Concentration=1500.0 (n=5)



SLS_Concentration=2000.0 (n=5)



SLS_Concentration=2500.0 (n=5)

